

Contexts and pseudocontexts

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Abstract

In the projection lattice of a finite-dimensional Hilbert space, an orthogonal basis – a *context* – is characterised by the property that its atomic probabilities sum to one on every state. We study a weakening of this condition, in which two tuples of atoms induce the same probability sum on every state. Such pairs are called *pseudocontexts*. In finite dimension this is equivalent to two distinct rank-one decompositions of the same positive operator, placing the notion naturally among finite frame-theoretic constructions. Size $k = 1$ is trivial; size $k = 2$ already admits genuinely nonorthogonal examples; and, in fact, examples exist for every $k \geq 2$. We present the operator formulation, give a general explicit construction, illustrate the symmetric $k = 3$ case with a figure, and list the remaining classification problems.

1 Introduction

Let L be the lattice of projections in a Hilbert space of finite dimension $n \geq 3$. Its atoms (minimal non-zero elements) correspond to one-dimensional subspaces, represented by nonzero vectors. States are described by probability measures $P: L \rightarrow [0, 1]$. We denote by $S(L)$ the set of all states on L .

Vectors $a_1, \dots, a_n$ form a *context* if they compose an orthogonal basis. In this case (and only in this case), they satisfy

$$\forall P \in S(L) : \sum_{i=1}^n P(a_i) = 1. \quad (1)$$

The present note concerns weaker additive identifications obtained by comparing two tuples of atoms that are not required to be orthogonal. Such *pseudocontexts* are quantum-logical shadows of orthomodular-lattice constructions of Mayet–Navara–Rogalewicz [3] and have direct physical meaning: they encode state-independent empirical equivalences between families of generally noncommuting observables. The answer is simple: pseudocontexts already exist for $k = 2$, and, more generally, for every $k \geq 2$.

2 Preliminaries

Throughout, H is a complex Hilbert space with $\dim H = n \geq 3$, and $L = L(H)$ is the orthomodular lattice of its closed subspaces. Its atoms correspond to one-dimensional subspaces, or equivalently to *rays* spanned by nonzero vectors. We identify an atom with any representative unit vector a and write $P_a = |a\rangle\langle a|$ for the orthogonal projection onto the ray it spans.

A *state* is a map $P: L \rightarrow [0, 1]$ with $P(I) = 1$ that is finitely additive on orthogonal families. By Gleason's theorem [1], every state arises from a density operator ρ via

$$P(a) = \text{tr}(\rho P_a), \quad \rho \geq 0, \text{ tr } \rho = 1. \quad (2)$$

The key consequence we use repeatedly is that two self-adjoint operators that agree on every state are equal as operators.

3 Contexts and pseudocontexts

The operator form of (1) is elementary and is the starting point for everything that follows.

Proposition 1 (Contexts). *Unit vectors $a_1, \dots, a_n \in H$ form an orthonormal basis if and only if*

$$\sum_{i=1}^n P_{a_i} = I. \quad (3)$$

Proof. The forward implication is immediate. For the converse, fix j and evaluate (3) on the vector a_j :

$$1 = \langle a_j, I a_j \rangle = \sum_{i=1}^n |\langle a_i, a_j \rangle|^2.$$

The term with $i = j$ already contributes 1, so all others must vanish. Hence $\langle a_i, a_j \rangle = 0$ for $i \neq j$. As this holds for every j , the family is orthonormal, and being of size n in an n -dimensional space it is an orthonormal basis. $\square$

Combined with (2), Proposition 1 recovers (1) as a state-theoretic characterisation of orthogonal bases.

Definition 2 (Pseudocontext). Let $k \geq 1$. A *pseudocontext of size k* is a pair of k -tuples of atoms $(a_1, \dots, a_k), (b_1, \dots, b_k)$ such that the underlying set $\{a_1, \dots, a_k, b_1, \dots, b_k\}$ has cardinality $2k$ and

$$\forall P \in S(L) : \sum_{i=1}^k P(a_i) = \sum_{i=1}^k P(b_i). \quad (4)$$

A pseudocontext is called *genuine* if neither k -tuple is a context.

By (2), condition (4) is equivalent to the operator identity

$$T := \sum_{i=1}^k P_{a_i} = \sum_{i=1}^k P_{b_i}, \quad (5)$$

which we call the *structure operator* of the pseudocontext. Thus pseudocontexts are precisely pairs of distinct rank-one decompositions of a single positive operator. This places them naturally among finite frame-theoretic constructions: contexts are the special case in which the structure operator is the identity on the relevant subspace; in general, pseudocontexts are rank-one decompositions of a common positive operator. If the common operator is proportional to the identity on its support, the family is a tight frame; after normalisation on that support, one obtains a POVM.

4 The smallest cases

Proposition 3 (Size $k = 1$). *There is no pseudocontext of size $k = 1$.*

Proof. The condition $P(a_1) = P(b_1)$ for all $P \in S(L)$ is equivalent, via (2), to $P_{a_1} = P_{b_1}$. This rank-one identity forces a_1 and b_1 to span the same ray, contradicting distinctness. $\square$

Corollary 4 (Size $k = 2$). *There are genuinely nonorthogonal pseudocontexts of size $k = 2$.*

Proof. Embed $\mathbb{C}^2$ into $\mathbb{C}^n$ as the span of the first two basis vectors and set

$$u_{\pm} = \left(\frac{\sqrt{3}}{2}, \pm \frac{1}{2}, 0, \dots, 0 \right), \quad v_{\pm} = \left(\frac{\sqrt{3}}{2}, \pm \frac{i}{2}, 0, \dots, 0 \right).$$

Then

$$P_{u_+} + P_{u_-} = P_{v_+} + P_{v_-} = \begin{pmatrix} 3/2 & 0 \\ 0 & 1/2 \end{pmatrix} \oplus 0_{n-2}.$$

The four rays are distinct, and within each pair the vectors are nonorthogonal. $\square$

Remark 5 (Real vs. Complex for $k = 2$). It is worth noting that genuinely nonorthogonal pseudocontexts of size $k = 2$ strictly require complex Hilbert spaces. In a real Hilbert space, the requirement that a trace-2 positive operator A admits a rank-one decomposition $A = P_c + P_d$ algebraically rigidly fixes the rays up to sign, forcing the two sets of rays to be identical. The availability of a continuous complex phase is what unlocks the $k = 2$ pseudocontexts in Corollary 4. For $k \geq 3$, however, pseudocontexts can be constructed entirely over the reals, as demonstrated in Section 6.

5 Pseudocontexts of every size $k \geq 2$

Theorem 6 (Complex-phase construction). *Let H contain an orthonormal pair e_1, e_2 . Fix an integer $k \geq 2$, let $\omega = e^{2\pi i/k}$, choose $\alpha \in (0, \pi/2)$, and choose $\phi \in \mathbb{R}$ such that $e^{i\phi}$ is not a k th root of unity. Define unit vectors*

$$a_j = \cos \alpha e_1 + \sin \alpha \omega^j e_2, \quad b_j = \cos \alpha e_1 + \sin \alpha e^{i\phi} \omega^j e_2, \quad j = 0, \dots, k-1.$$

Then the $2k$ rays spanned by the a_j and b_j are pairwise distinct, and

$$\sum_{j=0}^{k-1} P_{a_j} = \sum_{j=0}^{k-1} P_{b_j} = k(\cos^2 \alpha P_{e_1} + \sin^2 \alpha P_{e_2}).$$

Hence $(a_0, \dots, a_{k-1})$ and $(b_0, \dots, b_{k-1})$ form a pseudocontext of size k .

If $\alpha \neq \pi/4$, then the pseudocontext is genuine.

Proof. Write $c = \cos \alpha$ and $s = \sin \alpha$. For each j we have

$$P_{a_j} = c^2 P_{e_1} + s^2 P_{e_2} + cs(\omega^{-j}|e_1\rangle\langle e_2| + \omega^j|e_2\rangle\langle e_1|),$$

and

$$P_{b_j} = c^2 P_{e_1} + s^2 P_{e_2} + cs(e^{-i\phi}\omega^{-j}|e_1\rangle\langle e_2| + e^{i\phi}\omega^j|e_2\rangle\langle e_1|).$$

Summing over $j = 0, \dots, k-1$ and using

$$\sum_{j=0}^{k-1} \omega^j = 0$$

gives

$$\sum_{j=0}^{k-1} P_{a_j} = \sum_{j=0}^{k-1} P_{b_j} = k(c^2 P_{e_1} + s^2 P_{e_2}).$$

For distinctness, suppose $a_j = \lambda a_\ell$. Since $c \neq 0$, comparing the first coordinates gives $\lambda = 1$, and then $\omega^j = \omega^\ell$, so $j = \ell$. Similarly, if $a_j = \lambda b_\ell$, then $\lambda = 1$ and hence $\omega^j = e^{i\phi}\omega^\ell$, which is impossible by the choice of ϕ . The same argument shows that the b_j are pairwise distinct rays.

Finally,

$$\langle a_j, a_\ell \rangle = c^2 + s^2 \omega^{\ell-j}.$$

This can vanish only if $c^2 = s^2$ and $\omega^{\ell-j} = -1$, so if $\alpha \neq \pi/4$ then no two distinct rays in the same k -tuple are orthogonal. The same holds for the b_j . Hence the pseudocontext is genuine. $\square$

Remark 7. At the special value $\alpha = \pi/4$, the structure operator becomes

$$\frac{k}{2} I_{\text{span}\{e_1, e_2\}},$$

so the construction yields a unit-norm tight frame on the two-dimensional support. For $k = 2$ this is just a context; for $k \geq 3$ it gives a highly symmetric pseudocontext.

6 A symmetric $k = 3$ example

Surprisingly, the case $k = 3$ has a very transparent planar representation. It is the finite-dimensional analogue of the Mercedes-Benz frame.

Example 8 (Mercedes-Benz type pseudocontext). Fix a two-dimensional subspace $V \subset H$ with orthonormal basis (e_1, e_2) , and fix an angle

$$\theta \in (0, \pi) \setminus \{\pi/3, 2\pi/3\}.$$

Put, for $j = 0, 1, 2$,

$$a_j = \cos\left(\frac{2\pi j}{3}\right)e_1 + \sin\left(\frac{2\pi j}{3}\right)e_2, \quad b_j = \cos\left(\frac{2\pi j}{3} + \theta\right)e_1 + \sin\left(\frac{2\pi j}{3} + \theta\right)e_2.$$

Then the six rays are pairwise distinct, and

$$\sum_{j=0}^2 P_{a_j} = \sum_{j=0}^2 P_{b_j} = \frac{3}{2} I_V.$$

Thus $\{a_0, a_1, a_2\}$ and $\{b_0, b_1, b_2\}$ form a pseudocontext of size $k = 3$.

Proof. In the basis (e_1, e_2) , each projector has the form

$$P_{\cos \varphi e_1 + \sin \varphi e_2} = \begin{pmatrix} \cos^2 \varphi & \cos \varphi \sin \varphi \\ \cos \varphi \sin \varphi & \sin^2 \varphi \end{pmatrix}.$$

For $\varphi_j = 2\pi j/3$ and for $\varphi_j = \theta + 2\pi j/3$, the trigonometric identities

$$\sum_{j=0}^2 \cos(2\varphi_j) = 0, \quad \sum_{j=0}^2 \sin(2\varphi_j) = 0$$

imply that the off-diagonal terms cancel, while the diagonal terms each sum to $3/2$. Hence both sums equal $\frac{3}{2} I_V$. The exclusion $\theta \notin \{\pi/3, 2\pi/3\}$ guarantees that the two triples are distinct as sets of rays. $\square$

Remark 9 (Complex trine description). Example 8 has a compact complex analogue. Let $\omega = e^{2\pi i/3}$ and embed V into $\mathbb{C}^n$ as $\text{span}\{e_1, e_2\}$. The *trines*

$$u_j = \frac{1}{\sqrt{2}}(1, \omega^j, 0, \dots, 0), \quad v_j = \frac{1}{\sqrt{2}}(1, e^{i\phi} \omega^j, 0, \dots, 0), \quad j = 0, 1, 2,$$

yield the same phenomenon after a change of basis in V . The identity $1 + \omega + \omega^2 = 0$ makes the off-diagonal entries of $\sum_j P_{u_j}$ cancel, giving the common structure operator $\frac{3}{2} I_V$. This is the form most natural in frame-theoretic discussions and in connection with the Weyl–Heisenberg group.

Remark 10 (Nondegenerate structure operators). The trine family is highly symmetric: its structure operator is a scalar multiple of the identity on its support. More complicated examples with nondegenerate structure operators occur already in dimension three; see, for instance, the coordinatizations discussed in [4, 5]. Such examples are useful when one wants to move beyond purely symmetric frame orbits.

Similar constructions [3] were known when L was a more general orthomodular lattice [2]. Some of these lattices can be represented as subsets of a Hilbert lattice [4, 5].

7 Physical interpretation and constraints

A pseudocontext $(\{a_i\}, \{b_i\})$ implements an empirical equivalence between two families of rank-one observables $A_i = P_{a_i}$ and $B_i = P_{b_i}$. Although the operators within each family are generally noncommuting, the two linear functionals

$$\rho \mapsto \sum_i \text{tr}(\rho A_i), \quad \rho \mapsto \sum_i \text{tr}(\rho B_i)$$

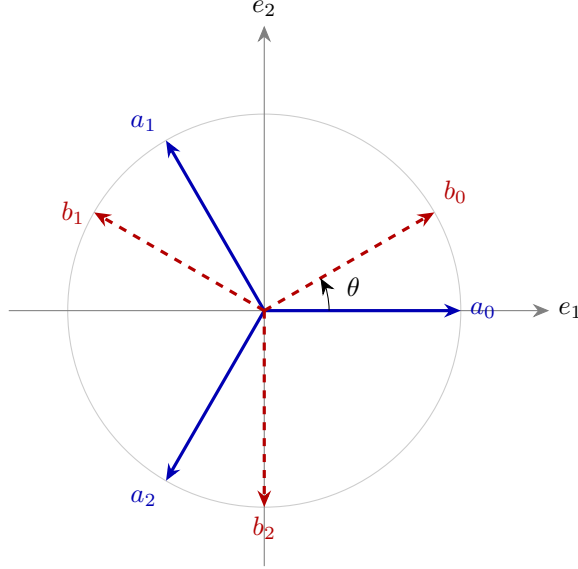


Figure 1: The Mercedes-Benz pseudocontext of Example 8, drawn for $\theta = \pi/6$. The solid blue rays a_0, a_1, a_2 and the dashed red rays b_0, b_1, b_2 are distinct unordered triples in the plane V , yet the associated rank-one projectors have the same sum $\frac{3}{2}I_V$.

agree on the whole state space. Because the structure operators are identical, this equivalence is strictly state-independent.

In the language of quantum information theory, the collections $\{A_i\}$ and $\{B_i\}$ are distinct rank-one operator decompositions of the same positive operator. If the common operator is proportional to the identity on its support, they form a tight frame; after normalisation on that support, one obtains a POVM. From a foundational standpoint, this is a state-independent additive equivalence closely related to, but distinct from, Kochen–Specker-type contextuality [6, 7].

8 Open problems

Having established the existence of pseudocontexts for every size $k \geq 2$ and isolated a simple $k = 3$ illustration, we highlight the remaining questions regarding their deeper classification:

Open Problem (Classification). Classify, up to unitary equivalence, all pseudocontexts of size k in $\mathbb{C}^n$. Even for small (k, n) , the orbit structure is not yet transparent.

Open Problem (Lattice representability). Characterise which orthomodular lattices carrying a pseudocontext-type additive identification admit a faithful embedding into some $L(\mathbb{C}^n)$, sharpening the partial results of [4, 5].

Open Problem (Frame-symmetry characterisation). Can one characterise pseudocontexts purely in terms of symmetries of the underlying rank-one frame — for example, in terms of unitaries commuting with the structure operator and permuting the atoms of the two decompositions?

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